

1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

```
PS C:\Users\heLL0> ping 192.168.17.1

Pinging 192.168.17.1 with 32 bytes of data:
Reply from 192.168.17.1: bytes=32 time<1ms TTL=128
Reply from 192.168.17.1: bytes=32 time<1ms TTL=128
Reply from 192.168.17.1: bytes=32 time<1ms TTL=128
Reply from 192.168.17.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.17.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

```
C:\Users\heLL0>tracert google.com

Tracing route to google.com [2404:6800:4007:808::200e]
over a maximum of 30 hops:

  0  5 ms  11 ms  5 ms  2401:4900:88b7:b5d5:a291:caff:fe28:3381
  1  *      *      *      Request timed out.
  2  28 ms  33 ms  37 ms  fc00:1::bb
  3  33 ms  23 ms  36 ms  2404:a800:1a00:805::11
  4  33 ms  188 ms  41 ms  2404:a800::167
  5  43 ms  38 ms  42 ms  2404:6800:8285:2c0::1
  6  35 ms  42 ms  39 ms  2404:6800:8285:2c0::1
  7  52 ms  43 ms  41 ms  2001:4860:0:1::2b4a
  8  45 ms  49 ms  41 ms  2001:4860:0:1::a858
  9  *      *      *      Request timed out.
 10  75 ms  77 ms  70 ms  2001:4860::9:4001:b923
 11  82 ms  91 ms  91 ms  2001:4860:0:1::67d
 12  77 ms  84 ms  229 ms  maa05s10-in-x0e.1e100.net [2404:6800:4007:808::200e]

Trace complete.
```

3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

```
C:\Users\hell0> ping 192.168.1.81

Pinging 192.168.1.81 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.81:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

```
C:\Users\hell0>arp -a

Interface: 192.168.1.88 --- 0xf
    Internet Address      Physical Address      Type
    192.168.1.1           a0-91-ca-28-33-81    dynamic
    192.168.1.81          58-ce-2a-12-81-3a    dynamic
    192.168.1.87          9c-da-3e-d5-ed-0a    dynamic
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.73.1 --- 0x11
    Internet Address      Physical Address      Type
    192.168.73.254        00-50-56-eb-12-37    dynamic
    192.168.73.255        ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.17.1 --- 0x17
    Internet Address      Physical Address      Type
    192.168.17.254        00-50-56-e9-0f-a5    dynamic
    192.168.17.255        ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Hint: Include screenshots or command outputs if possible.

Objective

The purpose of this practical was to understand how devices communicate over a local network using the **Ping**, **Traceroute**, and **ARP** commands. These tools help verify connectivity, identify network paths, and view connected devices.

Procedure

First, both the laptop and a mobile phone were connected to the same Wi-Fi network. The phone's IP address was identified through its Wi-Fi settings. The **ping** command was then used from the laptop to verify communication between the two devices. Successful replies confirmed proper network connectivity.

Next, the **tracert** www.youtube.com command was executed to observe the route taken by packets from the laptop to YouTube's servers. The command displayed multiple hops, showing the path through different routers before reaching the destination.

The phone was then disconnected from the Wi-Fi network, and the ping command was executed again. This time, the laptop displayed "**Request timed out**", indicating that the destination device was no longer reachable.

Finally, the **arp -a** command was used to list devices detected on the local network. The IP addresses were matched with known devices such as the router, laptop, mobile phone, and smart TV.